RFID SECURITY SYSTEM

Introduction

- Used Arduino IDE and Embedded C/C++ for programming, testing, and implementing an RFID-based security system.
- Designed and developed an RFID-based access control system for secure entry using Arduino and embedded C.
- Implemented unique ID verification logic to restrict unauthorized access.

Components

1. Arduino Uno - Main controller.
2. RFID Reader Module - Card reader
3. RFID Card - Access credential
4. LEDs (red and green) - Status indicators
5. Buzzer - Audio alert
6. USB Cable
7. Jumper Wires

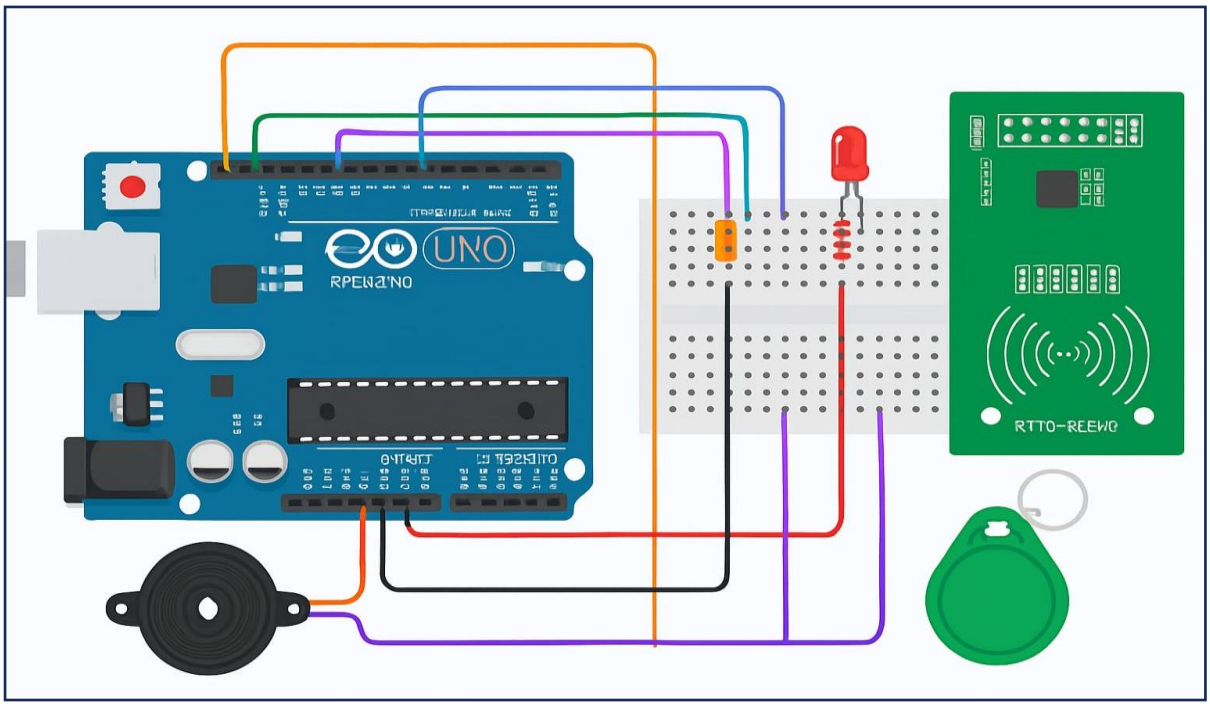
Tool

Arduino IDE

Programming Language

C programming

Circuit Diagram



Program

```
#include <SPI.h>
#include <MFRC522.h>

#define SS_PIN 10
#define RST_PIN 9
int redLed = 7;
int greenLed = 6;
int buzzer = 5;

MFRC522 mfrc522(SS_PIN, RST_PIN);   // Create MFRC522 instance.

void setup()
{
  pinMode(redLed, OUTPUT);
  pinMode(greenLed, OUTPUT);
```

```
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);   // Initiate a serial communication
  SPI.begin();      // Initiate  SPI bus
  mfrc522.PCD_Init();   // Initiate MFRC522
  Serial.println("Approximate your card to the reader...");
  Serial.println();

}
void loop()
{
  // Look for new cards
  if ( ! mfrc522.PICC_IsNewCardPresent())
  {
    return;
  }
  // Select one of the cards
  if ( ! mfrc522.PICC_ReadCardSerial())
  {
    return;
  }
  //Show UID on serial monitor
  Serial.print("UID tag :");
  String content= "";
  byte letter;
  for (byte i = 0; i < mfrc522.uid.size; i++)
  {
     Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
     Serial.print(mfrc522.uid.uidByte[i], HEX);
     content.concat(String(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " "));
     content.concat(String(mfrc522.uid.uidByte[i], HEX));
  }
  Serial.println();
  Serial.print("Message : ");
  content.toUpperCase();
  if (content.substring(1) == "2C 83 4C 4A") //Change the UID of the
card/cards that you want to give access
  {
    Serial.println("Authorized access");
    Serial.println();
    digitalWrite(greenLed, HIGH);

    digitalWrite(redLed, LOW);
    delay(3000);
```

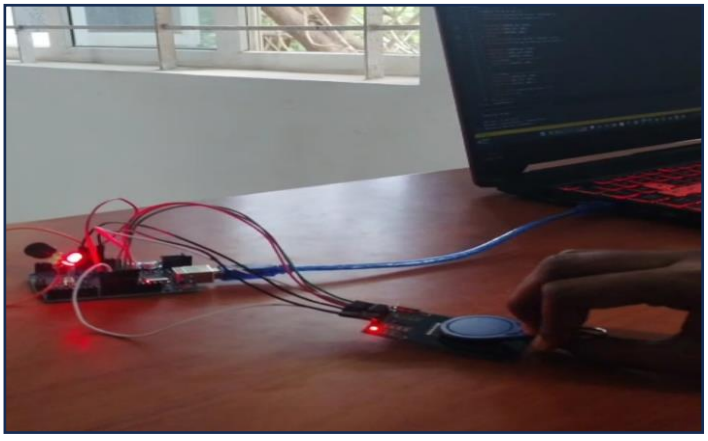
```
    digitalWrite(greenLed, LOW);
  }

 else   {
    Serial.println(" Access denied");
    digitalWrite(redLed, HIGH);
    digitalWrite(buzzer, HIGH);
    digitalWrite(greenLed, LOW);
    delay(3000);
    digitalWrite(buzzer, LOW);
    digitalWrite(redLed, LOW);

  }
}
```

Working Model





Working

- The Arduino Uno is powered through a USB cable connected to the laptop and supplies power to the connected components.
- When a user brings an RFID card near the RFID reader module, the reader scans the card and reads its unique identification number (UID).
- The RFID reader sends the scanned UID to the Arduino Uno through the connected communication pins.
- The Arduino compares the received UID with the authorized card IDs stored in its program.
- If the UID matches an authorized ID,
    - green LED - indicate that access is granted.
    - red LED and buzzer - indicate unauthorized access.
- After processing the card, the system returns to its initial state and waits for the next RFID card to be scanned.

Conclusion

- Authorized RFID cards are recognized, and access is granted.
- Unauthorized RFID cards are rejected, and an alert is generated using the buzzer.
- Red and green LEDs indicate access denied and access granted, respectively.